

DATA ANALYTICS REPORT

Customer Shopping Behavior Analysis

Trends, customer segments and strategic recommendations

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Python · SQL · Power BI

Scope

3,900 transactions · 18 variables · 4 product categories

EXECUTIVE SUMMARY

Key results at a glance

3,900 Customers analysed	\$59.76 Average purchase	\$233K Total revenue	3.75 / 5 Average review rating
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This report analyses 3,900 retail transactions to understand what drives spending, loyalty and subscription behaviour. The analysis combines Python (cleaning and feature engineering), PostgreSQL (ten business queries) and an interactive Power BI dashboard. The main conclusions are:

- **Revenue is driven by volume, not by basket size.** Male customers generate 67.7% of revenue because they account for 68% of purchases; average spend per purchase is almost identical across genders (\$59.54 vs. \$60.25).
- **Subscribers do not yet spend more.** Only 27% of customers subscribe, and their average spend (\$59.49) is in line with non-subscribers (\$59.87). Repeat buyers are not noticeably more likely to subscribe (27.6%).
- **The customer base is highly loyal but barely renewing.** 79.9% of customers fall in the “Loyal” segment, while only 83 customers (2.1%) are new.
- **Discounts are widespread.** 43% of purchases carry a discount and the five most discount-dependent products sit at 47–50%. Yet 839 customers still spend above average while using a discount.
- **Revenue is evenly spread across age groups** (from \$55.8K to \$62.1K), with Young Adults leading and Express shipping users spending about 3.5% more per order.

Bottom line: growth opportunities lie less in finding a “better” segment than in making subscriptions valuable, renewing the new-customer pipeline, and using discounts more selectively to protect margin.

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Business context and problem statement

A leading retail company wants to better understand its customers' shopping behaviour in order to improve sales, customer satisfaction and long-term loyalty. Management has noticed changes in purchasing patterns across demographics, product categories and sales channels (online vs. offline), and is particularly interested in which factors – discounts, reviews, seasons or payment preferences – drive consumer decisions and repeat purchases.

Central business question

"How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?"

Objectives

- Identify spending patterns across demographics, categories and shipping options.
- Segment customers by purchase history and measure loyalty and subscription behaviour.
- Assess the role of discounts and customer reviews in purchasing decisions.
- Translate the findings into concrete marketing, product and pricing recommendations.

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Data and methodology

Dataset overview

Attribute	Description
Volume	3,900 rows (one purchase per customer) × 18 columns
Customer demographics	Age, gender, location, subscription status
Purchase details	Item purchased, category, purchase amount (USD), season, size, colour
Shopping behaviour	Discount applied, promo code used, previous purchases, purchase frequency, review rating, shipping type, payment method
Data quality	37 missing values (0.9%), all in Review Rating

Data preparation (Python)

Step	Action
Loading & profiling	Dataset imported with pandas; structure and distributions checked with info() and describe() .
Missing values	The 37 missing review ratings were imputed with the median rating of each product category.
Standardisation	Column names converted to snake_case for readability and documentation.
Feature engineering	Created age_group (binned ages) and purchase_frequency_days .
Consistency check	promo_code_used was found redundant with discount_applied and was dropped.
Database load	Cleaned DataFrame loaded into PostgreSQL for SQL analysis.

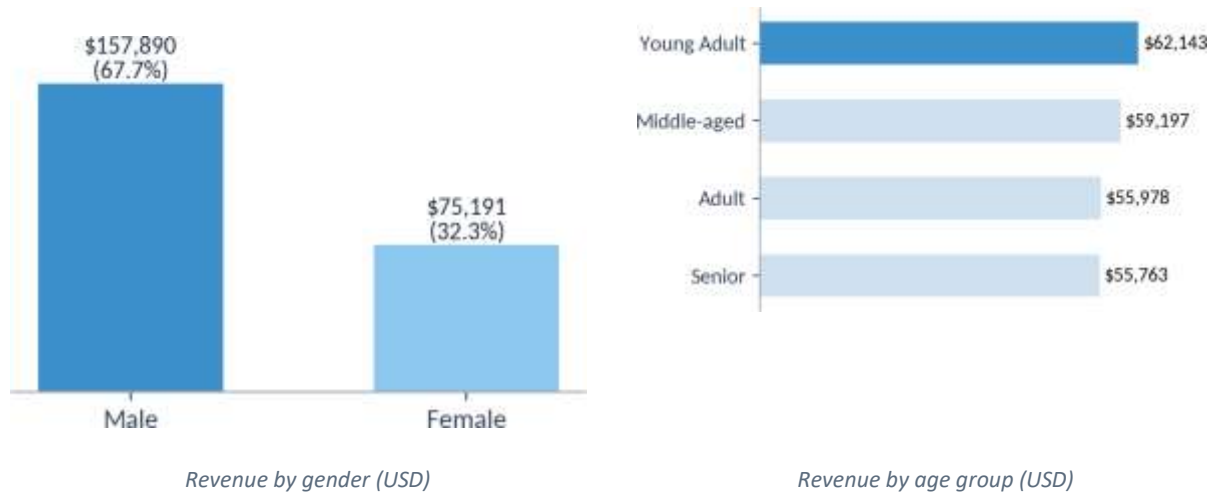
Analysis and reporting

Ten business questions were answered with SQL in PostgreSQL (section 3), then summarised in an interactive Power BI dashboard (section 4). Charts in this report are reproduced from the query results.

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Key findings

3.1 Revenue by gender



Male customers generated \$157,890 against \$75,191 for female customers. However, the dataset contains 2,652 male and 1,248 female purchases: average spend per purchase is virtually the same (\$59.54 vs. \$60.25). The gap is a **volume effect** – there are more male customers – not a spending effect.

3.2 Revenue by age group

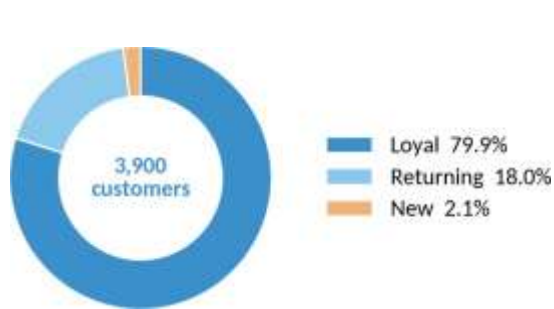
Young Adults contribute the most (\$62,143), followed by Middle-aged (\$59,197), Adult (\$55,978) and Senior (\$55,763) customers. The spread between the highest and lowest group is only 11%, so no age group can be neglected.

3.3 Subscribers vs. non-subscribers

Status	Customers	Share	Avg. spend	Total revenue
Subscribers	1,053	27.0%	\$59.49	\$62,645
Non-subscribers	2,847	73.0%	\$59.87	\$170,436

Subscription currently brings no spending premium. Among repeat buyers (more than five previous purchases), 958 subscribe and 2,518 do not (27.6% subscription rate, versus 27.0% overall): purchase frequency alone does not lead customers to subscribe, which leaves 2,518 loyal non-subscribers as a clear target pool.

3.4 Customer segmentation



Customer segments by purchase history



Most discount-dependent products (% of purchases discounted)

Loyal customers represent 3,116 of 3,900 (79.9%), Returning customers 701 (18.0%) and New customers only 83 (2.1%). A base this loyal is an asset, but the thin new-customer pipeline is a long-term risk and suggests the “Loyal” threshold may be too generous to discriminate between customers.

3.5 Discounts and shipping

Hats (50.00%), sneakers (49.66%), coats (49.07%), sweaters (48.17%) and pants (47.37%) are the most discount-dependent products. Despite discounts, 839 customers (21.5%) spent above the average purchase amount, showing that discounts do not necessarily cap basket value. Express shipping orders average \$60.48 versus \$58.46 for Standard (+3.5%).

3.6 Products and ratings

Category	#1	#2	#3
Accessories	Jewelry (171)	Sunglasses (161)	Belt (161)
Clothing	Blouse (171)	Pants (171)	Shirt (169)
Footwear	Sandals (160)	Shoes (150)	Sneakers (145)
Outerwear	Jacket (163)	Coat (161)	–

Top products per category by number of orders (shown values in brackets).

The five best-rated products are Gloves (3.86), Sandals (3.84), Boots (3.82), Hat (3.80) and Skirt (3.78). Ratings are tightly clustered around the overall average of 3.75, so reviews alone differentiate products only marginally. Clothing is the largest category with 1,737 purchases (44.5% of the total).

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Power BI dashboard

An interactive dashboard allows business teams to explore the data by subscription status, gender, category and shipping type. It tracks three headline KPIs – 3.9K customers, \$59.76 average purchase amount and 3.75 average review rating – alongside revenue and sales breakdowns by category and age group.



Customer Behavior Dashboard (Power BI)

How to use it

- Filter by **subscription status** to compare subscriber and non-subscriber contribution.
- Use the **category** and **shipping type** slicers to see how revenue and sales volume shift.
- Compare the **age group** panels to spot where revenue and order volume diverge.

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Business recommendations

Priority	Recommendation	Evidence	Suggested action / KPI
1	Make subscriptions worth it	27% subscribe; no spend uplift; 2,518 repeat buyers not subscribed	Design exclusive benefits (early access, free shipping thresholds); target repeat non-subscribers. KPI: subscription rate, spend gap
2	Refine loyalty programme	79.9% already “Loyal”; only 2.1% new	Introduce tiers within “Loyal”; reward repeat buyers; add acquisition campaigns. KPI: new-customer share, tier mix
3	Review discount policy	43% of purchases discounted; top items at 47–50%	Test reduced discount depth on hats, sneakers, coats; keep targeted offers for price-sensitive items. KPI: margin, conversion
4	Sharpen product positioning	Ratings 3.78–3.86; Clothing = 44.5% of purchases	Feature top-rated and best-selling items in campaigns; improve low-rated lines. KPI: category sales, rating
5	Targeted marketing	Young Adults highest revenue; Express +3.5% spend	Prioritise Young Adult and Middle-aged segments; promote Express shipping; broaden appeal to female customers. KPI: revenue per segment

Limitations and next steps

The business problem also refers to online vs. offline channels, seasons and payment preferences. The current analysis does not yet cover these dimensions, although the data (season, payment method, review rating) is available:

- Analyse sales by **season** and **payment method** to guide campaign timing and checkout options.
- Measure the relationship between **review rating** and **repeat purchase** to quantify the effect of satisfaction on loyalty.
- Add margin and channel data to evaluate discounts on profitability, not only on volume.
- Document the thresholds behind the New / Returning / Loyal segments and test alternative definitions.